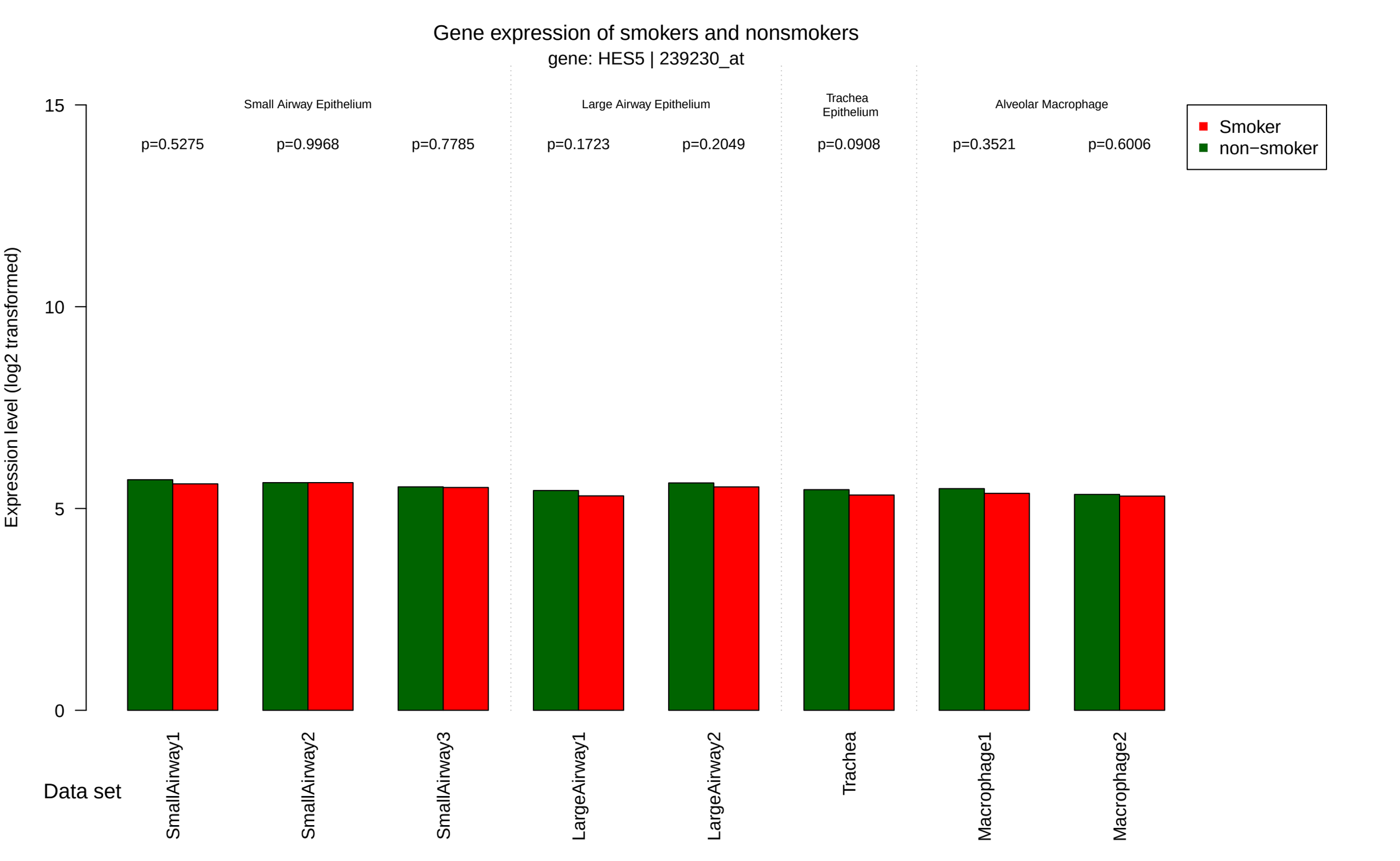
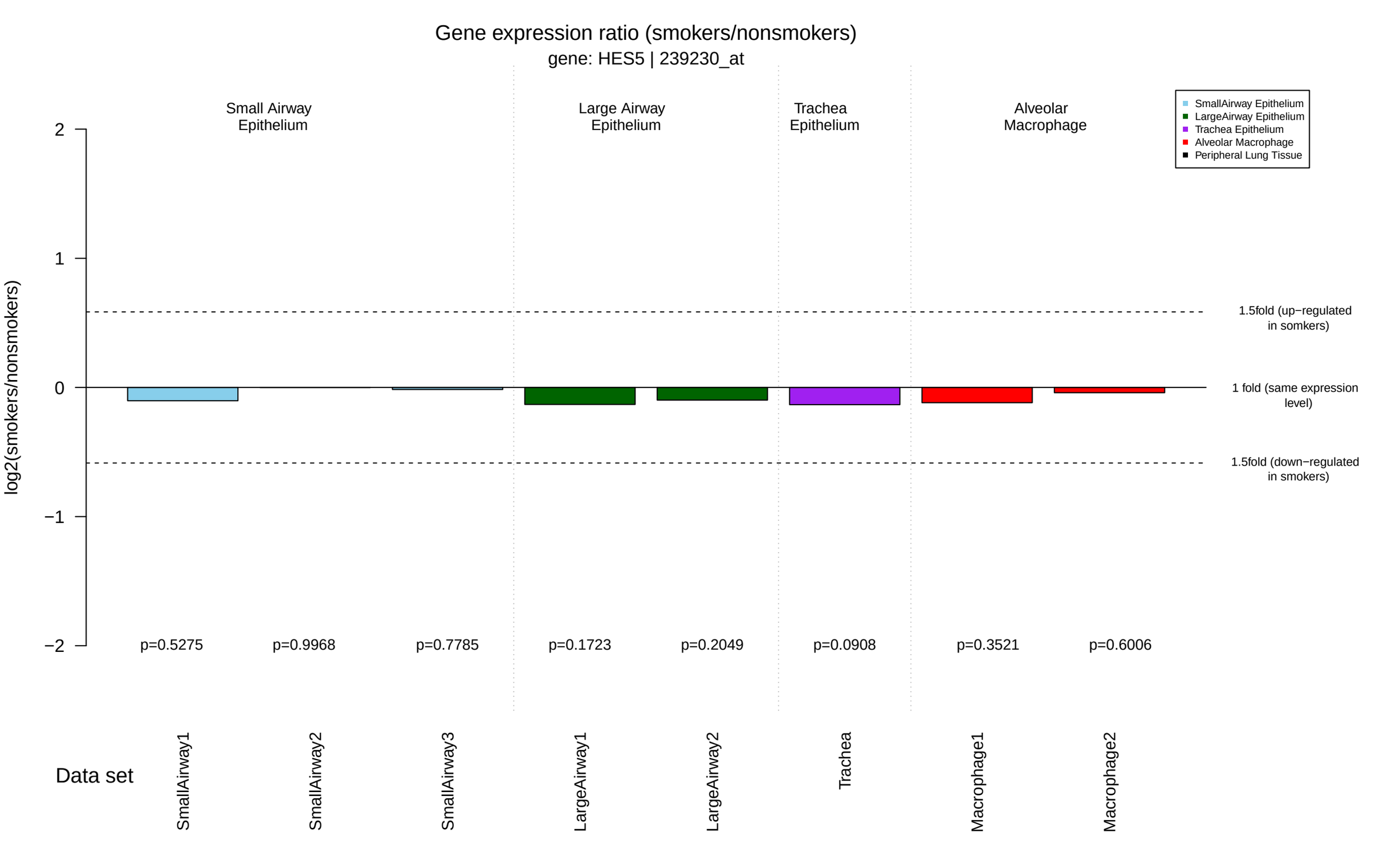
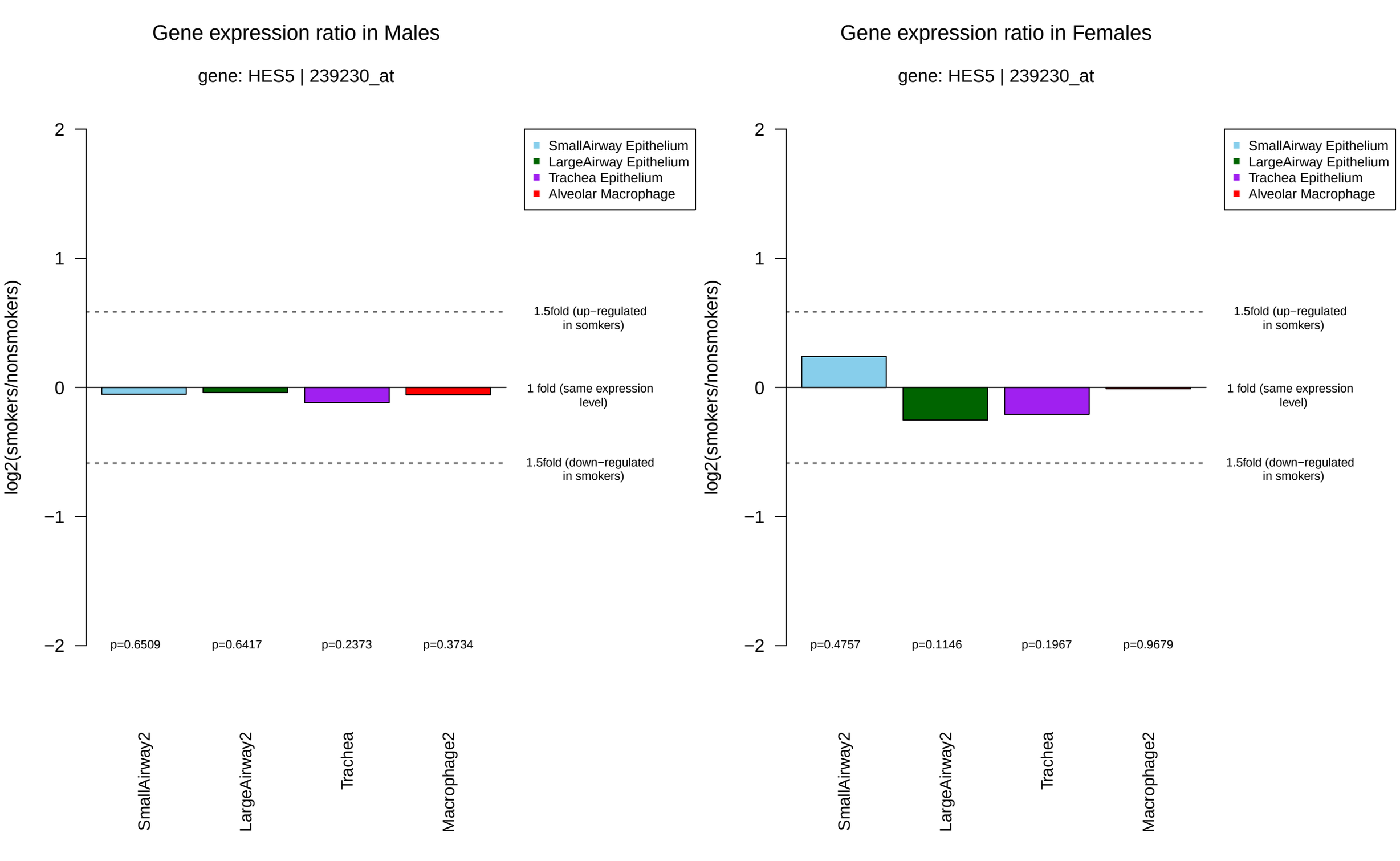
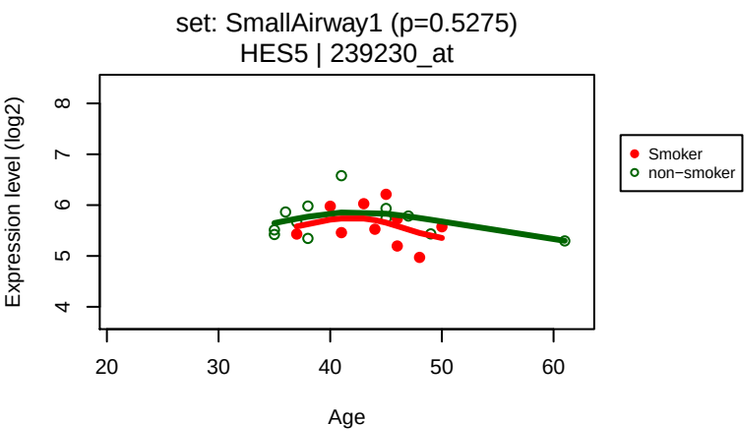
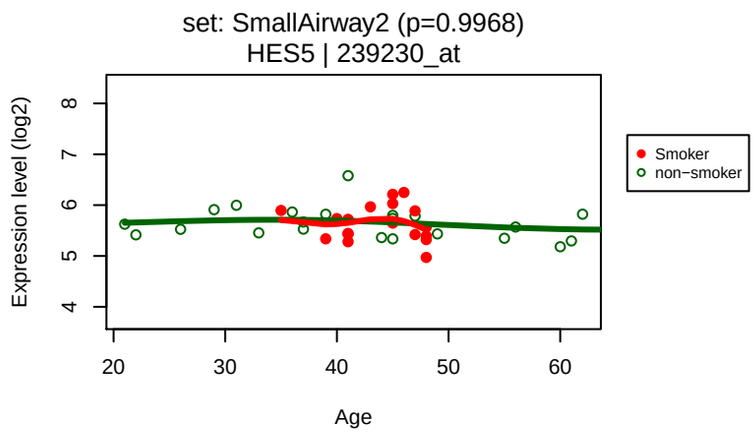
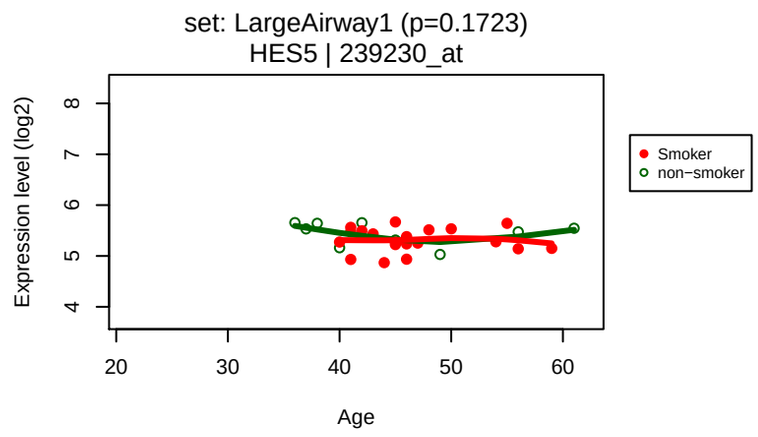

Gene expression of smokers and nonsmokers
gene: HES5 | 239230_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5275
p=0.9968
p=0.7785
p=0.1723
p=0.2049
p=0.0908
p=0.3521
p=0.6006
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

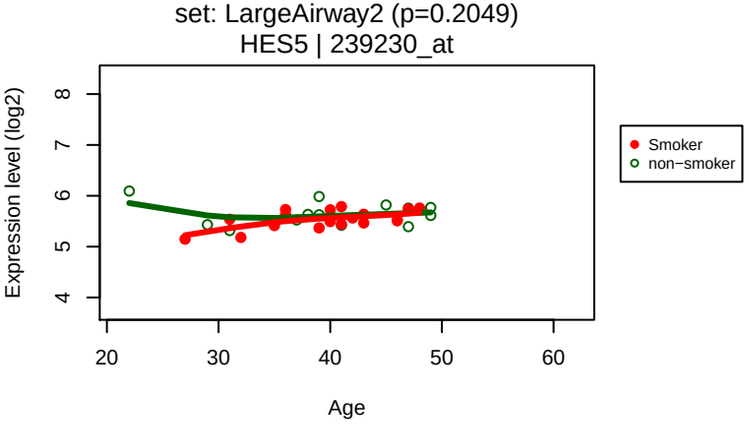

Gene expression ratio (smokers/nonsmokers)
gene: HES5 | 239230_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5275
p=0.9968
p=0.7785
p=0.1723
p=0.2049
p=0.0908
p=0.3521
p=0.6006
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

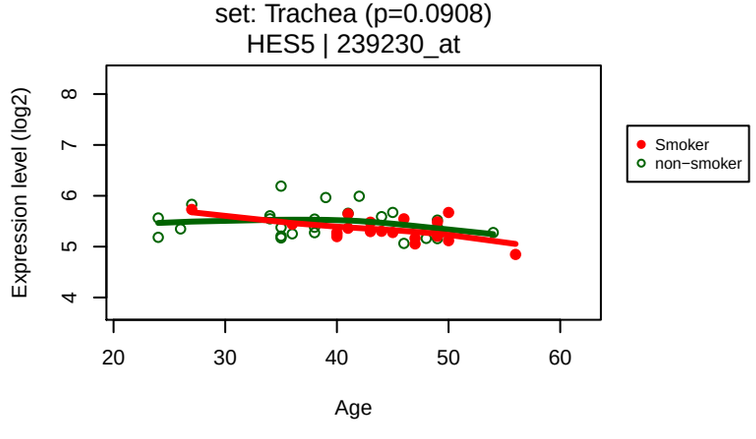

Gene expression ratio in Males
gene: HES5 | 239230_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6509
p=0.6417
p=0.2373
p=0.3734
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HES5 | 239230_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4757
p=0.1146
p=0.1967
p=0.9679
SmallAirway2
LargeAirway2
Trachea
Macrophage2

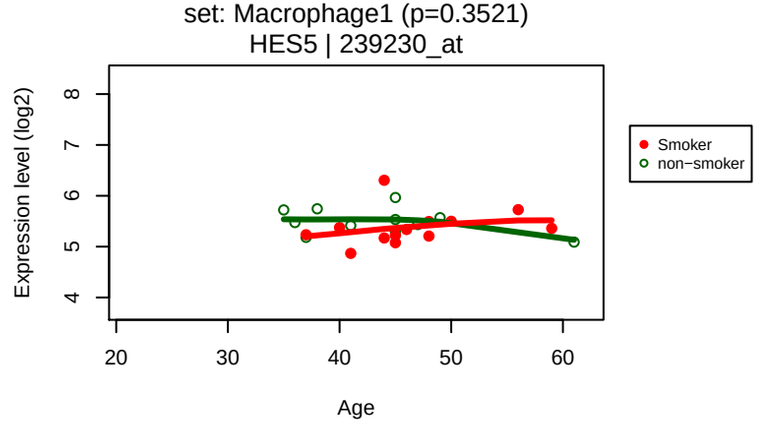

set: SmallAirway1 (p=0.5275)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker

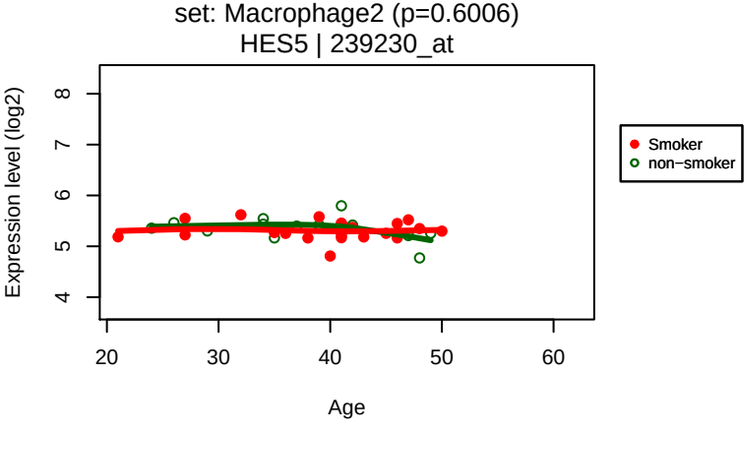

set: SmallAirway2 (p=0.9968)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker

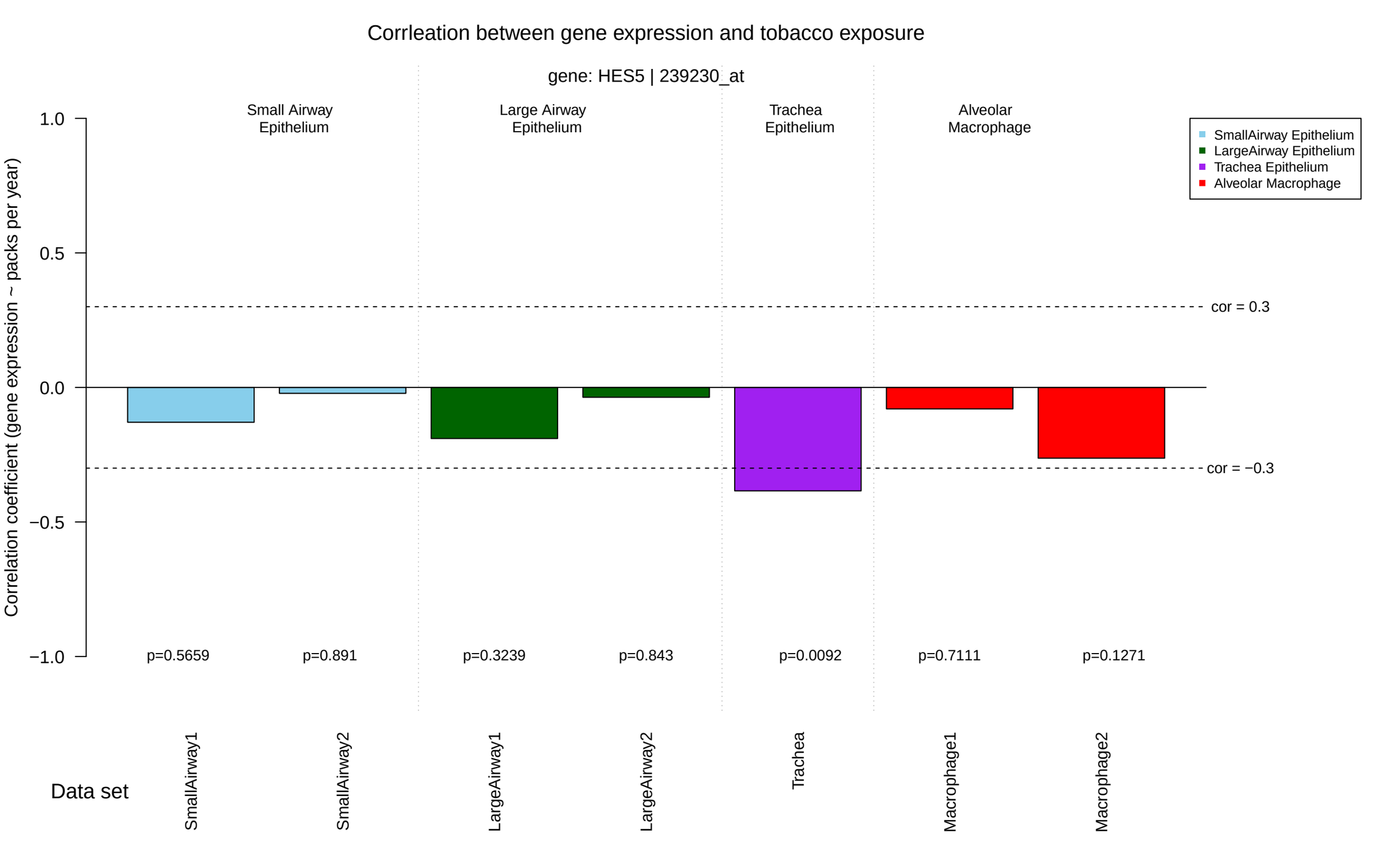

set: LargeAirway1 (p=0.1723)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2049)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0908)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3521)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6006)
HES5 | 239230_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HES5 | 239230_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5659
p=0.891
p=0.3239
p=0.843
p=0.0092
p=0.7111
p=0.1271
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2